

# Listings: Exploratory Data Analysis Report

This report will observe the Listing dataset and prepare it for data analysis.

## Dataset Overview

```
Rows in Listing Dataset: 930311
```

```
Columns in Listing Dataset: 84
```

```
Index(['OriginalListPrice', 'ListingKey', 'ListAgentEmail', 'CloseDate',
      'ClosePrice', 'ListAgentFirstName', 'ListAgentLastName', 'Latitude',
      'Longitude', 'UnparsedAddress', 'PropertyType', 'LivingArea',
      'ListPrice', 'DaysOnMarket', 'ListOfficeName', 'BuyerOfficeName',
      'CoListOfficeName', 'ListAgentFullName', 'CoListAgentFirstName',
      'CoListAgentLastName', 'BuyerAgentMlsId', 'BuyerAgentFirstName',
      'BuyerAgentLastName', 'FireplacesTotal', 'AssociationFeeFrequency',
      'AboveGradeFinishedArea', 'ListingKeyNumeric', 'MLSAreaMajor',
      'TaxAnnualAmount', 'CountyOrParish', 'PropertyType.1', 'MlsStatus',
      'ElementarySchool', 'ListAgentFirstName.1', 'AttachedGarageYN',
      'ParkingTotal', 'BuilderName', 'PropertySubType', 'LotSizeAcres',
      'SubdivisionName', 'BuyerOfficeAOR', 'YearBuilt', 'DaysOnMarket.1',
      'BuyerAgencyCompensationType', 'StreetNumberNumeric', 'LivingArea.1',
      'ListingId', 'BathroomsTotalInteger', 'City', 'BuyerAgencyCompensation',
      'TaxYear', 'BuildingAreaTotal', 'BedroomsTotal',
      'ContractStatusChangeDate', 'Longitude.1', 'ElementarySchoolDistrict',
      'CoBuyerAgentFirstName', 'PurchaseContractDate', 'ListingContractDate',
      'BelowGradeFinishedArea', 'BusinessType', 'Latitude.1', 'ListPrice.1',
      'StateOrProvince', 'CoveredSpaces', 'MiddleOrJuniorSchool',
      'FireplaceYN', 'Stories', 'HighSchool', 'Levels', 'ListAgentLastName.1',
      'CloseDate.1', 'LotSizeDimensions', 'LotSizeArea', 'MainLevelBedrooms',
      'NewConstructionYN', 'GarageSpaces', 'HighSchoolDistrict', 'PostalCode',
      'BuyerOfficeName.1', 'AssociationFee', 'LotSizeSquareFeet',
      'MiddleOrJuniorSchoolDistrict', 'UnparsedAddress.1'],
      dtype='object')
```

Important note: While loading the monthly CSV files into a Pandas dataframe, several columns ran into a DtypeWarning.

```
/var/folders/86/cdz_5pvx705dzpfgv_0qsc600000qn/T/ipykernel_25030/143335703.py:17:
DtypeWarning: Columns (32,58,63,66) have mixed types. Specify dtype option on import
or set low_memory=False.
  listing = pd.read_csv(listing_filename)
```

Columns with mixed values are defaulted to object type. However, it is important to understand the expected data types of each column and make adjustments accordingly. Since there are 84 columns, it is efficient to perform this after feature selection (or when each feature is analyzed).

## Data Filtering

Observe the following property types:

```
array(['ManufacturedInPark', 'CommercialSale', 'Residential',  
      'ResidentialLease', 'Land', 'ResidentialIncome', 'CommercialLease',  
      'BusinessOpportunity'], dtype=object)
```

And filter to Residential.

```
Listing rows before filtering: 930311  
Listing rows after filtering: 591980
```

## Data Cleaning

Validate completeness

Out of the 82 columns, the following are missing more than 90% of values:

|                              | Missing Count | Missing Percent |
|------------------------------|---------------|-----------------|
| FireplacesTotal              | 591980        | 100.00          |
| AboveGradeFinishedArea       | 591980        | 100.00          |
| TaxAnnualAmount              | 591980        | 100.00          |
| BuilderName                  | 564621        | 95.38           |
| TaxYear                      | 591980        | 100.00          |
| BuildingAreaTotal            | 539108        | 91.07           |
| ElementarySchoolDistrict     | 591980        | 100.00          |
| CoBuyerAgentFirstName        | 577610        | 97.57           |
| BelowGradeFinishedArea       | 588608        | 99.43           |
| BusinessType                 | 591980        | 100.00          |
| CoveredSpaces                | 591980        | 100.00          |
| LotSizeDimensions            | 560960        | 94.76           |
| MiddleOrJuniorSchoolDistrict | 591980        | 100.00          |

Based on the above table, all but BuildingAreaTotal can also be dropped. Many of these features are categorized as metadata, meaning they don't hold meaningful contributions to the property value overall. Furthermore, columns with 100% missing values don't have any contribution to analysis. BuildingAreaTotal can be retained as it is considered a market analysis feature and has the potential to provide meaningful contributions to analysis.

\* LotSizeDimensions is dropped because the dataset already records LotSizeAcre and LotSizeSquareFeet.

## Key Numeric Fields

Analyze the distribution of key numeric fields: ClosePrice, ListPrice, OriginalListPrice, LivingArea, LotSizeAcres, BedroomsTotal, BathroomsTotalInteger, DaysOnMarket, and YearBuilt.

For each field, generate histograms, boxplots, and percentile summaries, and identify extreme outliers for later handling. Ensure that the features are their expected data type prior.

```
ClosePrice          float64
ListPrice           float64
OriginalListPrice   float64
LivingArea          float64
LotSizeAcres        float64
BedroomsTotal       float64
BathroomsTotalInteger float64
DaysOnMarket        int64
YearBuilt           float64
dtype: object
```

The features do not require changes to data type.

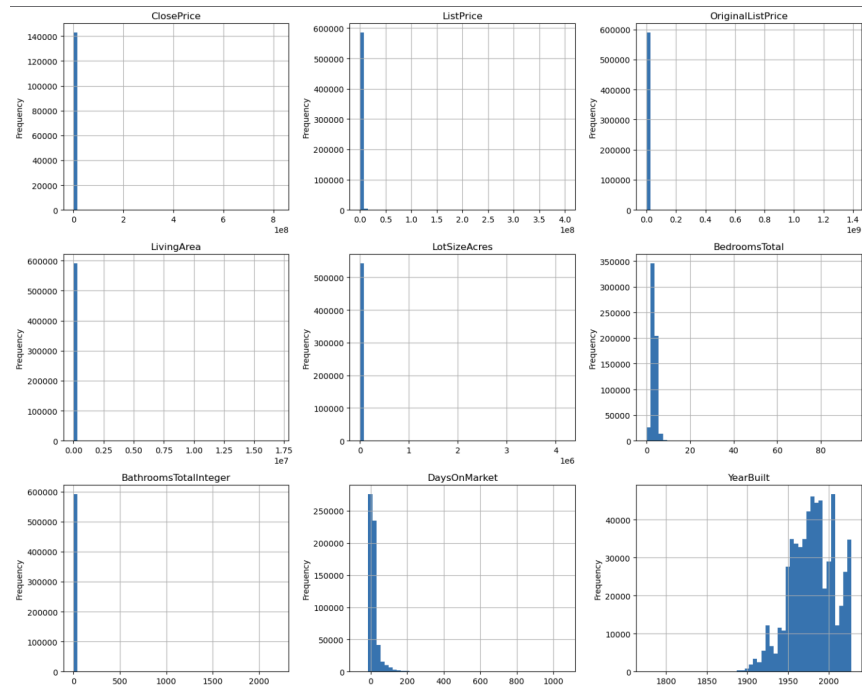
## Percentile Summaries

|       | ClosePrice   | ListPrice    | OriginalListPrice | LivingArea   | LotSizeAcres | BedroomsTotal | BathroomsTotalInteger | DaysOnMarket  | YearBuilt     |
|-------|--------------|--------------|-------------------|--------------|--------------|---------------|-----------------------|---------------|---------------|
| count | 1.431820e+05 | 5.919800e+05 | 5.911370e+05      | 5.913800e+05 | 5.433390e+05 | 591818.000000 | 591923.000000         | 591980.000000 | 590963.000000 |
| mean  | 1.209867e+06 | 1.318847e+06 | 1.402616e+06      | 1.980178e+03 | 6.138693e+01 | 3.226718      | 2.630373              | 18.463484     | 1979.690109   |
| std   | 4.332265e+06 | 2.410601e+06 | 7.310655e+06      | 2.234135e+04 | 1.159600e+04 | 1.188212      | 3.134990              | 25.805270     | 27.027556     |
| min   | 5.250000e+02 | 1.000000e+00 | 0.000000e+00      | 0.000000e+00 | 0.000000e+00 | 0.000000      | 0.000000              | -58.000000    | 1776.000000   |
| 25%   | 6.000000e+05 | 5.820000e+05 | 5.850000e+05      | 1.248000e+03 | 1.200000e-01 | 2.000000      | 2.000000              | 5.000000      | 1961.000000   |
| 50%   | 8.550000e+05 | 8.490000e+05 | 8.490000e+05      | 1.672000e+03 | 1.700000e-01 | 3.000000      | 2.000000              | 11.000000     | 1980.000000   |
| 75%   | 1.350839e+06 | 1.385000e+06 | 1.395000e+06      | 2.302000e+03 | 3.161000e-01 | 4.000000      | 3.000000              | 21.000000     | 2001.000000   |
| max   | 8.200000e+08 | 4.000000e+08 | 1.390000e+09      | 1.702132e+07 | 4.187292e+06 | 94.000000     | 2208.000000           | 1063.000000   | 2028.000000   |

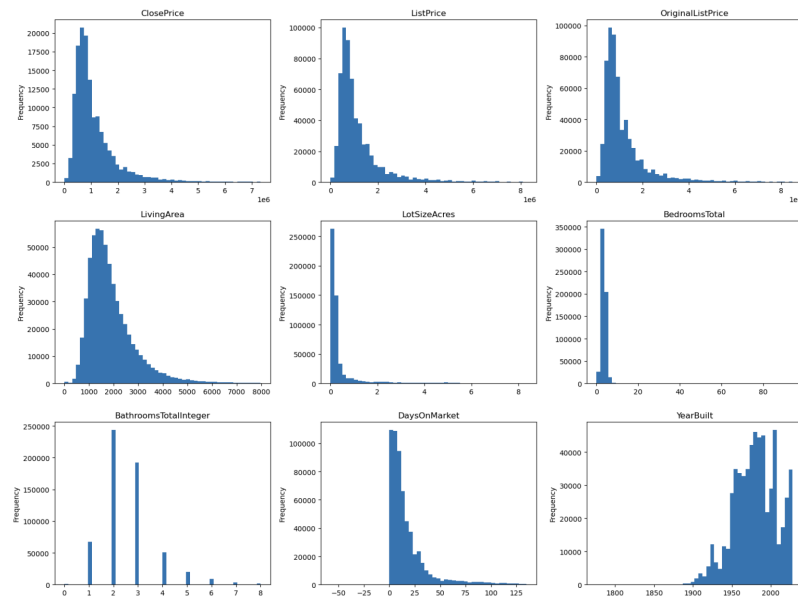
## Extreme Outliers

|   | Feature               | Q1        | Q3         | IQR       | Lower Bound | Upper Bound | Outlier Count | Outlier Percent |
|---|-----------------------|-----------|------------|-----------|-------------|-------------|---------------|-----------------|
| 0 | ClosePrice            | 600000.00 | 1350838.75 | 750838.75 | -526258.12  | 2477096.88  | 10533         | 1.78            |
| 1 | ListPrice             | 582000.00 | 1385000.00 | 803000.00 | -622500.00  | 2589500.00  | 49814         | 8.41            |
| 2 | OriginalListPrice     | 585000.00 | 1395000.00 | 810000.00 | -630000.00  | 2610000.00  | 48980         | 8.27            |
| 3 | LivingArea            | 1248.00   | 2302.00    | 1054.00   | -333.00     | 3883.00     | 29271         | 4.94            |
| 4 | LotSizeAcres          | 0.12      | 0.32       | 0.20      | -0.17       | 0.61        | 87055         | 14.71           |
| 5 | BedroomsTotal         | 2.00      | 4.00       | 2.00      | -1.00       | 7.00        | 1923          | 0.32            |
| 6 | BathroomsTotalInteger | 2.00      | 3.00       | 1.00      | 0.50        | 4.50        | 37676         | 6.36            |
| 7 | DaysOnMarket          | 5.00      | 21.00      | 16.00     | -19.00      | 45.00       | 49124         | 8.30            |
| 8 | YearBuilt             | 1961.00   | 2001.00    | 40.00     | 1901.00     | 2061.00     | 1491          | 0.25            |

# Histograms



Removing the extreme outliers from plots:



# Boxplots

